core MPU, the power requirement increases as per its cores.

So, calculate how much power it is going to consume and check whether it is permissible or not. Perform heat dissipation analysis of the electronics and analyse how much temperature rise will take place inside the enclosure. This step is very crucial while designing the hardware.

Less power demand and less heat dissipation are the main advantages of an efficient power supply. Heat plays a major role in the longevity of electronic components, particularly capacitors. When exposed to heat, their lifespan starts decreasing.

Memory requirements

After fulfilling the power supply needs, you can go for microcontroller selection. The MIPS rating and power supply consumption are some of the important things that one should track.

Another important thing for microcontroller selection is the memory requirement. The memory contains static RAM (scratch pad memory for software execution) and program memory (flash memory or ROM for storing software programs). So, the question is how much RAM and program memory is adequate?

It is difficult to analyse this question and arrive at an absolute conclusion at the beginning because a software/product regularly undergoes upgradation and customisation whenever new features are demanded by the industry. You also need to keep some space for peripheral obsolescence management. For instance, if the peripheral components like an A-D converter go obsolete, then they will need to be replaced. That may require more software drivers to be added in the future. Additions and changes in communication protocols and standards will require firmware upgrades in future. The new software versions and drivers keep on getting more resource hungry as the memory requirement keeps increasing.

For a typical embedded system not having a heavy-duty operating system like Linux or a simple task scheduler kind of an operating system, you can consider doubling the program memory size than your usual requirement. Also, double the RAM or static memory size as per your requirement. As far as processing speed is concerned, you can consider twice the capacity of your requirement. Any situation can arise later on like fixing a major bug, change in electronics or addition of a feature. Your MPU and memory should be able to accommodate such software changes.

If you wish to sell a product commercially, then first start with cost versus feature analysis. But if you wish to create a unique technology, then perform a proof of concept (PoC). For instance, when thinking of designing a vibration sensing IoT gateway or device for the first time, one approach is to buy a development kit or ready-to-use hardware, which is not going to be your end-product but just experimental hardware.

Else, you can build your hardware. Try the technology and get hands-on experience before optimising it.

Electronics considerations (analogue and digital)

After selecting the processors, you can then turn to analogue or digital electronics requirements. For this, you may need to use op-amps. Before selecting them, several parameters need to be considered. If your application demands a higher frequency, then choose the ones with better gain bandwidth. If you are amplifying a precision signal from sensors that give a millivolt or microvolt output, then you need precision and low-drift op-amps whose temperature drift is minimal.

Take care of the PCB routing and follow the EMI/EMC guidelines while routing PCBs for all your circuits. If there are analogue circuits, make a board of multiple layers having a dedicated ground layer for better EMI/EMC and ground current handling. Avoid ground current loops on your PCB. With the help of simulation, you can test your analogue electronics first before including it in a circuit diagram for PCB layout.

Takeaway

With plenty of talented engineers graduating in India, the future is bright for start-ups in embedded hardware design. The new generation of designers is getting smarter with the help of a lot of online content available to study.

Indian start-ups should keep the international focus, and not just stick to Indian requirements of quality standards. Go for a product that can be sold outside the Indian market, whether in Europe or America. Or at least aim to be up to that standard.

Do not skip EMI/EMC compliances while making the product and strictly abide by the safety standards, be it in India or abroad.

If the Indian start-ups compromise on quality and make cheap, low-cost products, then there will always be a gap between the Indian and foreign-made products. Indian products should be at par with European and American products.

If you follow the steps and envisage the changes that are going to come along your hardware design journey, then your path is simplified. The flow of hardware design will be smooth if each step is taken care of. EFY

The article is based on the talk 'Simplifying Embedded Hardware Design' by Mahendra Karandikar, GM - Developments and Projects at Jay Instruments & Systems, which was presented during January edition of Tech World Congress 2021